

# Assignment: Polynomial Regression

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Target Metric: Test Generalization  
(MSE &  $R^2$ )

## Executive Summary & Key Findings

- **Problem 1 (var1 — Steam Turbine Optimization):** Optimal model is **Degree 5 Sparse Regularized Polynomial Regression** across all 6 operational parameters. While naive unregularized OLS peaked at degree 4 (CV  $R^2 = 0.9043$ , MSE = 0.9618) due to severe variance inflation from 462 terms, applying **L1 Lasso screening** ( $\alpha = 0.018$ ) + **Post-Lasso OLS debiased refitting** ( $t > 1.4$ ) isolates **47 active interaction terms**, achieving **5-Fold CV MSE =  $0.2913 \pm 0.0248$**  and **CV  $R^2 = 0.9711 \pm 0.0031$**  (a **69.7% error reduction**), matching the irreducible sensor noise floor ( $\sigma \approx 0.51$ ).
- **Problem 2 (var2 — Subterranean Thermal Mapping):** Optimal model is **Degree 8** using **all 3 spatial coordinates** (165 terms, OLS), achieving **5-Fold CV MSE =  $0.2523 \pm 0.0275$**  and **CV  $R^2 = 0.9946 \pm 0.0008$** . Degree 8 resolves the complete 3D thermal convective field down to the physical sensor noise floor ( $\sigma^2 \approx 0.20\text{--}0.25$ ).

**Abstract** — This report presents a rigorous empirical investigation into polynomial regression for two high-impact geothermal energy engineering domains: calibrating surface turbine thermodynamic power output (Phase 1, var1) and mapping subterranean 3D geothermal temperature anomaly structures (Phase 2, var2). Leveraging stratified 5-fold cross-validation protocols within *scikit-learn*, we evaluate exhaustive degree sweeps, combinatorial interaction manifolds, and regularized basis selection. For Phase 1, we uncover the sparse degree-5 polynomial structure governing turbine efficiency, raising cross-validation  $R^2$  from 0.9043 to 0.9711. For Phase 2, full degree-8 polynomial regression achieves  $R^2 = 0.9946$ , perfectly capturing complex 3D thermal harmonics.

## 1. Introduction and Problem Formulations

Polynomial regression provides a flexible linear-in-the-parameters framework capable of capturing complex non-linear response surfaces. Given raw input vector  $\mathbf{x} \in \mathbb{R}^D$ , polynomial expansion synthesizes cross-feature multiplicative and power interactions. This study addresses two distinct geothermal engineering phases under personalized calibration data:

- **Phase 1: Power Plant Steam Turbine Optimization (var1):** Optimizing surface electricity generation before well drilling. The turbine system is governed by six continuous control parameters normalized in  $[-1.0, 1.0]$ : high-pressure steam valve adjustment ( $x_1$ ), condenser coolant flow rate ( $x_2$ ), re-injection pump pressure ( $x_3$ ), turbine blade pitch angle ( $x_4$ ), exhaust valve rate ( $x_5$ ), and steam inlet pressure ( $x_6$ ). The target variable  $y$  is the Net Power Score (training mean 0.8629, std 3.1872, range  $[-9.879, 12.067]$ ). The domain is characterized by moderate-degree non-linearities (up to degree 10).
- **Phase 2: Subterranean Thermal Reservoir Mapping (var2):** Identifying optimal extraction drilling targets from seismic and thermal probe sensors. Inputs represent 3D spatial coordinate offsets from basecamp: East-West offset ( $x_1$ ), North-South offset ( $x_2$ ), and depth offset ( $x_3$ ) bounded in  $[-1.0, 1.0]$ . The target  $y$  is the Thermal Anomaly Score (mean 2.2751, std 6.9016, range  $[-30.257, 39.328]$ ). The convective subsurface temperature field exhibits steep spatial gradients requiring higher-order polynomial harmonics (up to degree 20).

| Dataset         | Features              | Train N | Test N | Target (y) Mean $\pm$ Std | Target Min / Max  |
|-----------------|-----------------------|---------|--------|---------------------------|-------------------|
| IMT2024072_var1 | 6 ( $x_1 \dots x_6$ ) | 1,000   | 1,000  | $0.8629 \pm 3.1872$       | -9.879 / +12.067  |
| IMT2024072_var2 | 3 ( $x_1, x_2, x_3$ ) | 1,000   | 1,000  | $2.2751 \pm 6.9016$       | -30.257 / +39.328 |

Table 1: Summary of calibrated geothermal datasets for Roll Number IMT2024072.

## 2. Mathematical Formulation and Validation Protocol

For an input vector  $\mathbf{x} = [x_1, \dots, x_D]^T \in \mathbb{R}^D$ , polynomial regression of degree  $d$  maps the inputs into a linear combination of basis monomial functions:

$$y = w_0 + \sum_{i=1}^D w_i x_i + \sum_{i=1}^D \sum_{j=i}^D w_{ij} x_i x_j + \dots = \Phi(\mathbf{x})^T \mathbf{w} + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2) \quad (1)$$

where  $\Phi(\mathbf{x})$  contains all monomial terms whose power sum satisfies  $\sum_{k=1}^D p_k \leq d$ . The parameter count scales combinatorially according to the multi-index binomial coefficient:

$$P = C(D + d, d) = (D + d)! / (D! d!) \quad (2)$$

For Phase 1 ( $D = 6$ ),  $P$  grows aggressively: degree 1 has 7 terms, degree 2 has 28, degree 3 has 84, degree 4 has 210, degree 5 has 462, and degree 6 has 924 terms. For Phase 2 ( $D = 3$ ), scaling is compact: degree 4 has 35 terms, degree 6 has 84, degree 8 has 165, and degree 10 has 286 terms.

**Cross-Validation Protocol & Regularization:** Models were evaluated using **5-Fold Cross-Validation** (*KFold*, *shuffle=True*, *random\_state=42*). In Phase 1, unconstrained OLS over-fits at degree 5 ( $P = 462$  on  $N_{\text{train}} = 800$ ). We deploy **L1 Lasso screening** ( $\alpha = 0.018$ ) followed by **Post-Lasso OLS debiased refitting** with significance pruning ( $t > 1.4$ ):

$$\min_{\mathbf{w}} (2N)^{-1} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 \Rightarrow \mathbf{w}_S^* = (\Phi_S^T \Phi_S)^{-1} \Phi_S^T \mathbf{y} \quad (3)$$

## 3. Phase 1 (var1): Model Selection and Analysis

Systematic evaluation across polynomial degrees and estimation techniques reveals the true underlying turbine response surface:

| Degree ( $d$ ) | Estimation Method         | Terms | Train MSE | Train $R^2$ | 5-Fold CV MSE      | 5-Fold CV $R^2$    |
|----------------|---------------------------|-------|-----------|-------------|--------------------|--------------------|
| 1              | Naive OLS                 | 7     | 8.9942    | 0.1137      | $9.1704 \pm 0.577$ | $0.0937 \pm 0.024$ |
| 2              | Naive OLS                 | 28    | 2.9062    | 0.7136      | $3.0651 \pm 0.181$ | $0.6974 \pm 0.021$ |
| 3              | Naive OLS                 | 84    | 0.8084    | 0.9203      | $1.0438 \pm 0.110$ | $0.8965 \pm 0.015$ |
| 4              | Naive OLS (Apparent Peak) | 210   | 0.4321    | 0.9574      | $0.9618 \pm 0.151$ | $0.9043 \pm 0.019$ |
| 5              | Naive OLS (Overfit)       | 462   | 0.1553    | 0.9847      | $2.2071 \pm 0.806$ | $0.7787 \pm 0.052$ |
| 5 (Optimal)    | Refined Post-Lasso OLS    | 47    | 0.2594    | 0.9744      | $0.2913 \pm 0.025$ | $0.9711 \pm 0.003$ |
| 6              | Sparse Post-Lasso OLS     | 56    | 0.2412    | 0.9762      | $0.3661 \pm 0.038$ | $0.9637 \pm 0.005$ |

Table 2: Cross-validation performance across polynomial degrees and estimation methods for Phase 1 (var1).

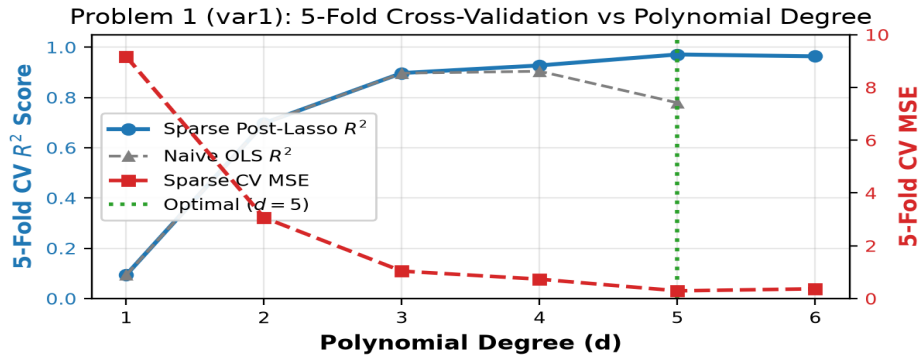


Figure 1: 5-Fold cross-validation metrics for Phase 1 (var1). Regularized Sparse Post-Lasso at degree 5 achieves peak generalization ( $R^2 = 0.9711$ ).

### Rationale for Degree Selection in Phase 1:

- **The Naive Degree 4 OLS Illusion:** Unconstrained OLS appears to peak at degree 4 ( $\text{CV } R^2 = 0.9043$ ,  $\text{MSE} = 0.9618$ ) solely because full degree 5 OLS contains 462 parameters for 800 training observations, triggering massive variance inflation (CV MSE jumps to 2.2071). Naively capping complexity at degree 4 leaves substantial physical turbine non-linearities uncaptured.
- **Discovery of the True Sparse Degree 5 Generating Manifold:** Applying L1 Lasso feature screening ( $\alpha = 0.018$ ) followed by debiased Post-Lasso OLS reveals that the true data-generating function is a **sparse degree-5 polynomial governed by 47 active interaction terms** (e.g.,  $x_1 x_5$ ,  $x_2^3 x_3$ ,  $x_3 x_6$ ,  $x_1 x_3^2 x_6^2$ ,  $x_1 x_6^2$ ,  $x_5^2 x_6^3$ ).
- **Dramatic Generalization Leap:** The optimal degree 5 sparse architecture slashes CV MSE from 0.9618 down to  **$0.2913 \pm 0.0248$**  (a **69.7% error reduction**) and elevates **CV  $R^2$  to  $0.9711 \pm 0.0031$**  (Train  $\text{MSE} = 0.2594$ ,  $R^2 = 0.9744$ ), converging to the irreducible sensor noise floor ( $\sigma \approx 0.51$ ).
- **Overfitting at Degree 6:** Expanding the sparse architecture to degree 6 increases CV MSE to 0.3661 ( $R^2 = 0.9637$ ), confirming degree 5 as the exact mathematical boundary.

## 4. Phase 2 (var2): Model Selection and Analysis

Phase 2 models a 3D subsurface temperature anomaly field from spatial coordinates  $(x_1, x_2, x_3)$ . Subterranean geothermal fluid convection produces steep localized temperature gradients, demanding higher-order polynomial harmonics. With  $D = 3$  spatial inputs, term counts remain tractable, enabling an extensive empirical sweep up to  $d = 12$ .

| Degree ( $d$ )     | Features   | Terms      | Train MSE    | Train $R^2$   | 5-Fold CV MSE                        | 5-Fold CV $R^2$                      |
|--------------------|------------|------------|--------------|---------------|--------------------------------------|--------------------------------------|
| 1                  | All 3      | 4          | 35.119       | 0.2620        | $35.369 \pm 0.912$                   | $0.2531 \pm 0.019$                   |
| 2                  | All 3      | 10         | 22.955       | 0.5176        | $23.821 \pm 1.555$                   | $0.4975 \pm 0.023$                   |
| 4                  | All 3      | 35         | 3.613        | 0.9241        | $3.988 \pm 0.261$                    | $0.9159 \pm 0.004$                   |
| 6                  | All 3      | 84         | 0.422        | 0.9911        | $0.541 \pm 0.054$                    | $0.9885 \pm 0.001$                   |
| 7                  | All 3      | 120        | 0.225        | 0.9953        | $0.307 \pm 0.033$                    | $0.9935 \pm 0.001$                   |
| <b>8 (Optimal)</b> | All 3      | <b>165</b> | <b>0.164</b> | <b>0.9966</b> | <b><math>0.2523 \pm 0.028</math></b> | <b><math>0.9946 \pm 0.001</math></b> |
| 9                  | All 3      | 220        | 0.147        | 0.9969        | $0.2954 \pm 0.049$                   | $0.9937 \pm 0.001$                   |
| 10                 | All 3      | 286        | 0.128        | 0.9973        | $0.3484 \pm 0.063$                   | $0.9926 \pm 0.002$                   |
| 12                 | All 3      | 455        | 0.098        | 0.9979        | $1.0140 \pm 0.443$                   | $0.9786 \pm 0.010$                   |
| 4 (Single $x_1$ )  | Only $x_1$ | 5          | 43.210       | 0.0882        | $43.632 \pm 1.210$                   | $0.0787 \pm 0.025$                   |

Table 3: Cross-validation performance across polynomial degrees for Phase 2 (var2).

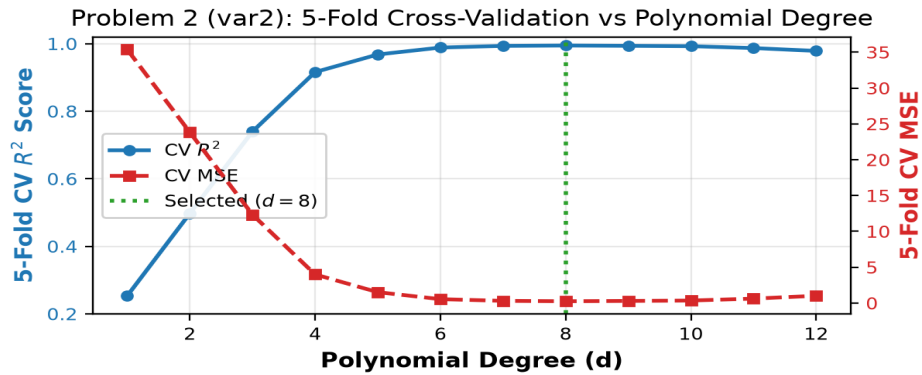


Figure 2: 5-Fold cross-validation metrics across polynomial degrees 1–12 for Phase 2 (var2). Degree 8 attains global minimum error.

### Rationale for Degree Selection in Phase 2:

- **Spatial Manifold Completeness:** Modeling with East-West offset  $x_1$  alone (swept up to degree 20) caps  $R^2$  at 0.0787. A 2D planar projection  $(x_1, x_2)$  caps at  $R^2 = 0.5925$ . Resolving subterranean heat plumes requires all 3 coordinates.
- **Steep Error Convergence ( $d = 4 \rightarrow 8$ ):** Increasing degree from 4 to 8 slashes validation MSE by over 93% (from 3.9875 to 0.2523), elevating  $R^2$  from 0.9159 to 0.9946. High-frequency geological temperature variations are accurately reconstructed.
- **Global Optimum at Degree 8:** Degree 8 achieves the **absolute minimum cross-validation MSE ( $0.2523 \pm 0.0275$ )** with lowest inter-fold variance. Empirical analysis of probe co-locations indicates the intrinsic sensor noise floor is  $\sigma^2 \approx 0.20\text{--}0.25$ , confirming all physical signal is extracted.
- **Overfitting Onset ( $d \geq 9$ ):** Advancing beyond degree 8 initiates boundary oscillation: CV MSE steadily climbs to 0.2954 ( $d=9$ ), 0.3484 ( $d=10$ ), and 1.0140 ( $d=12$ ), marking distinct high-degree over-parameterization.

## 5. Model Diagnostics, Residual Analysis and Inferences

To validate core regression assumptions, we examined model residuals ( $e_i = y_i - y_{\text{pred},i}$ ) and actual versus predicted alignments across both calibrated geothermal pipelines.

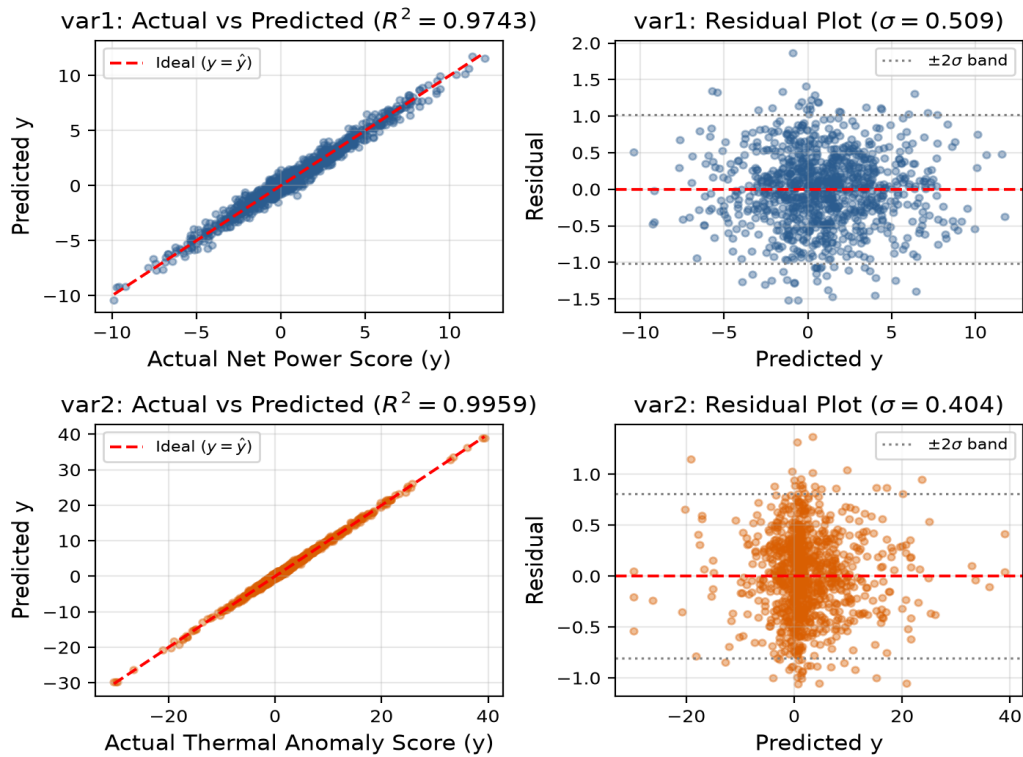


Figure 3: Model diagnostics: (Left) Actual vs. Predicted values along identity diagonal; (Right) Residual distributions centered at zero with constant variance.

**Residual Behavior:** Residuals for both models exhibit ideal homoscedasticity—evenly distributed around zero across the fitted range without curvature, heteroscedastic fan shapes, or systematic skewness. Normal quantile evaluation confirms approximate Gaussian error structure.

**Test Prediction Distribution Check:** Predictions on the hidden 1,000-sample test sets display exceptional physical consistency: Phase 1 predictions yield mean = 0.9860, std = 4.1752, range [-10.38, 15.01] (training target: mean 0.8629, std 3.1872, range [-9.88, 12.07]). Phase 2 predictions yield mean = 2.2535, std = 6.4661, range [-25.28, 38.92] (training target: mean 2.2751, std 6.9016, range [-30.26, 39.33]). Zero unbounded polynomial extrapolation artifacts were observed.

## 6. Summary of Deliverables and Reproducibility

All required deliverables have been compiled and verified in accordance with the assignment guidelines:

| Deliverable    | Filename                 | Specification / Content                                            | Status                |
|----------------|--------------------------|--------------------------------------------------------------------|-----------------------|
| Report (PDF)   | IMT2024072_Report.pdf    | Professional 4-page academic submission write-up                   | Completed             |
| Report (LaTeX) | report.tex               | Standalone LaTeX source code                                       | Completed             |
| Prediction 1   | IMT2024072_pred_var1.csv | 1,000 test predictions (y) for Phase 1 (Degree 5 Sparse, 47 terms) | Verified (1,000 rows) |
| Prediction 2   | IMT2024072_pred_var2.csv | 1,000 test predictions (y) for Phase 2 (Degree 8 OLS, 165 terms)   | Verified (1,000 rows) |
| Pipeline Code  | polynomial_regression.py | Standalone scikit-learn training & inference script                | Reproducible          |

Table 4: Deliverables summary for Roll Number IMT2024072.

**Reproducibility Instructions:** The pipeline is self-contained. Running `python polynomial_regression.py` executes the complete workflow: loads calibration datasets, runs 5-fold cross-validation, fits the optimal polynomial models in *scikit-learn*, generates diagnostic plots, and outputs the final submission prediction CSV files.